



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 01

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Program:

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
arr = [10, 25, 30, 45, 50]  
key = 30  
result = linear_search(arr, key)  
if result != -1:  
    print("Key found at index", result)  
else:  
    print("Key not found")
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def linear_search(arr, key): 2 for i in range(len(arr)): 3 if arr[i] == key: 4 return i 5 return -1 6 arr = [10, 25, 30, 45, 50] 7 key = 30 8 result = linear_search(arr, key) 9 if result != -1: 10 print("Key found at index", result) 11 else: 12 print("Key not found")</pre>			Key found at index 2